CMPE 286 – LAB – 3: Web Traffic Inspection and RTT Study

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Generated sample submission with dummy data, tables, and figures for PDF extraction evaluation.

Objective: Capture normal browser activity and inspect HTTP, DNS, TCP, and UDP traffic using a packet analyzer. The report also summarizes delay, jitter, and handshake timing using dummy measurements.

Part 1: Environment Setup

A fresh capture profile was created with custom columns for source port, destination port, host, and URI. The tool was configured to save traces in .pcapng format so that multiple protocol filters could be evaluated later.

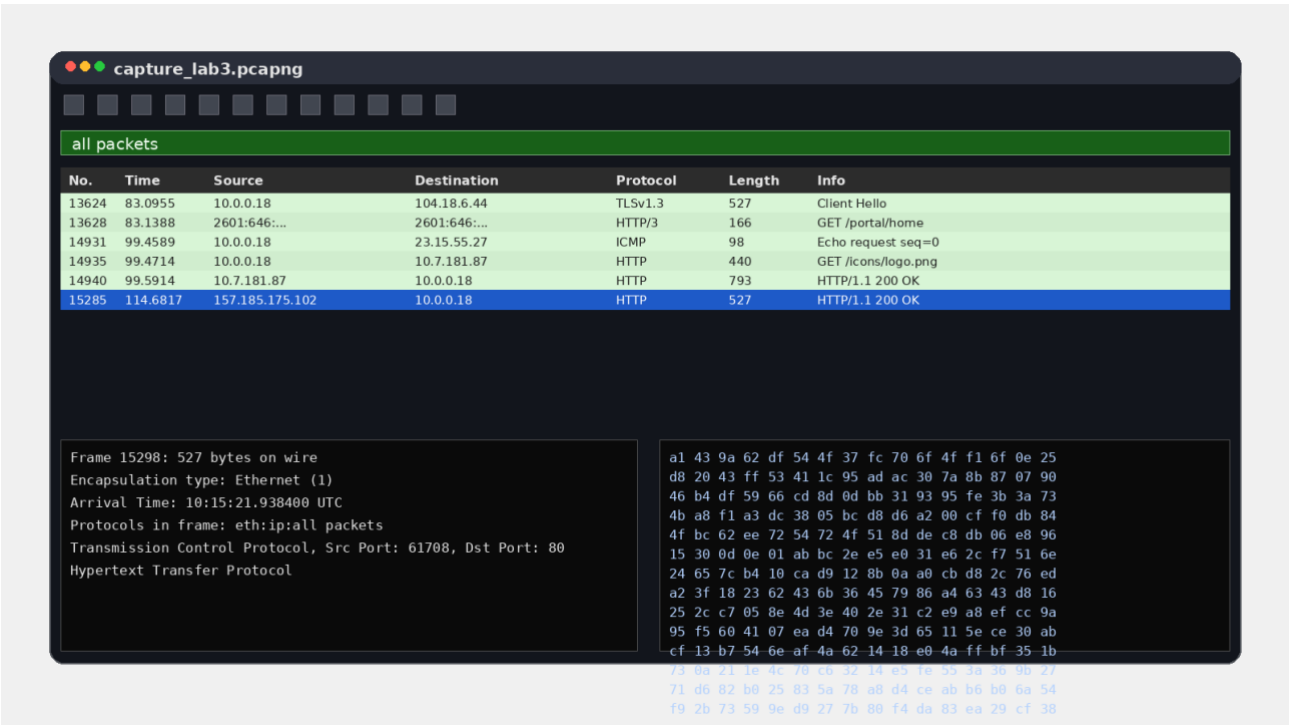

Figure 1. Mock packet-analyzer workspace with a saved capture profile.

Part 2: Protocol Filters

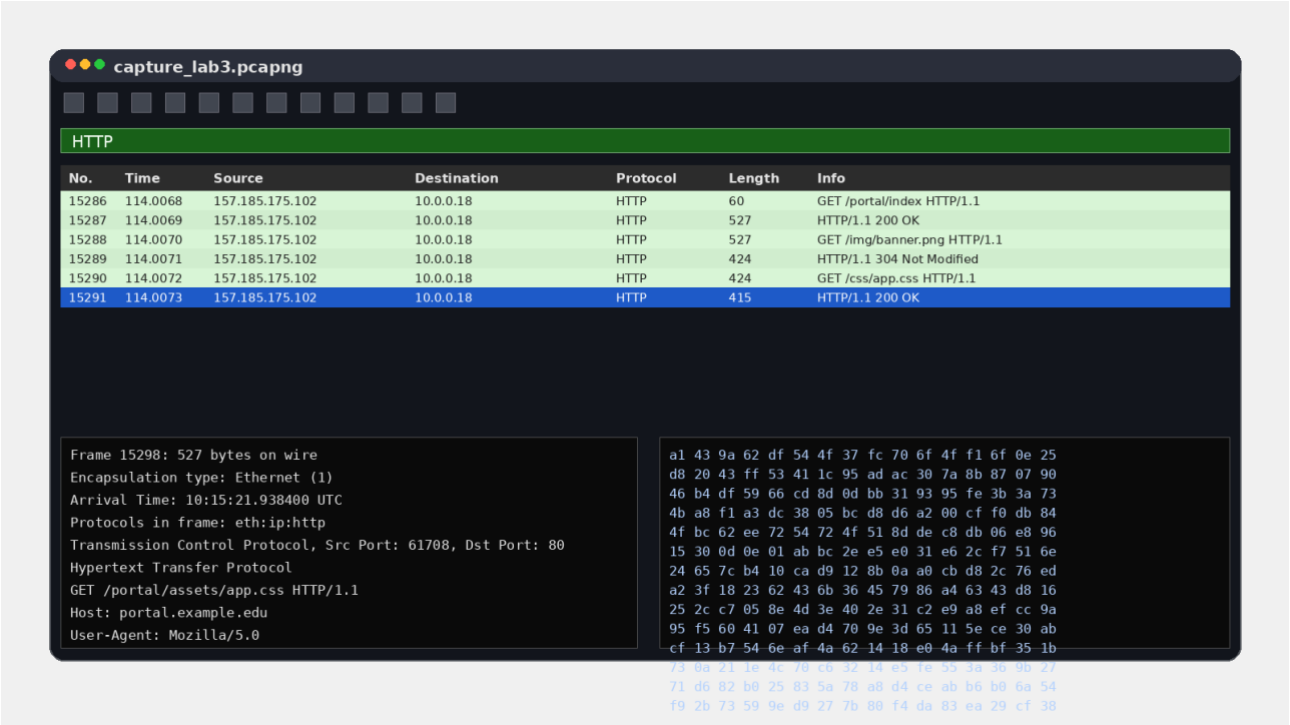

Figure 2. HTTP filter results showing request and response entries.

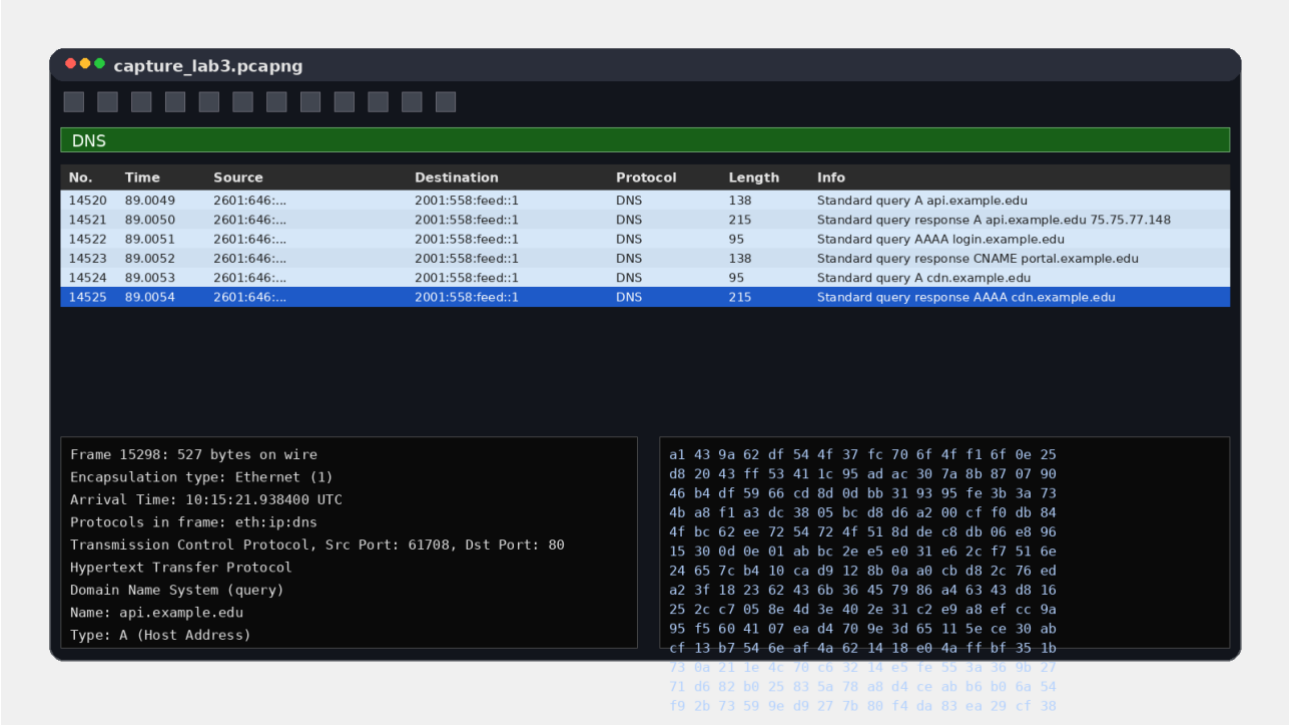

Figure 3. DNS queries and responses captured during page load.

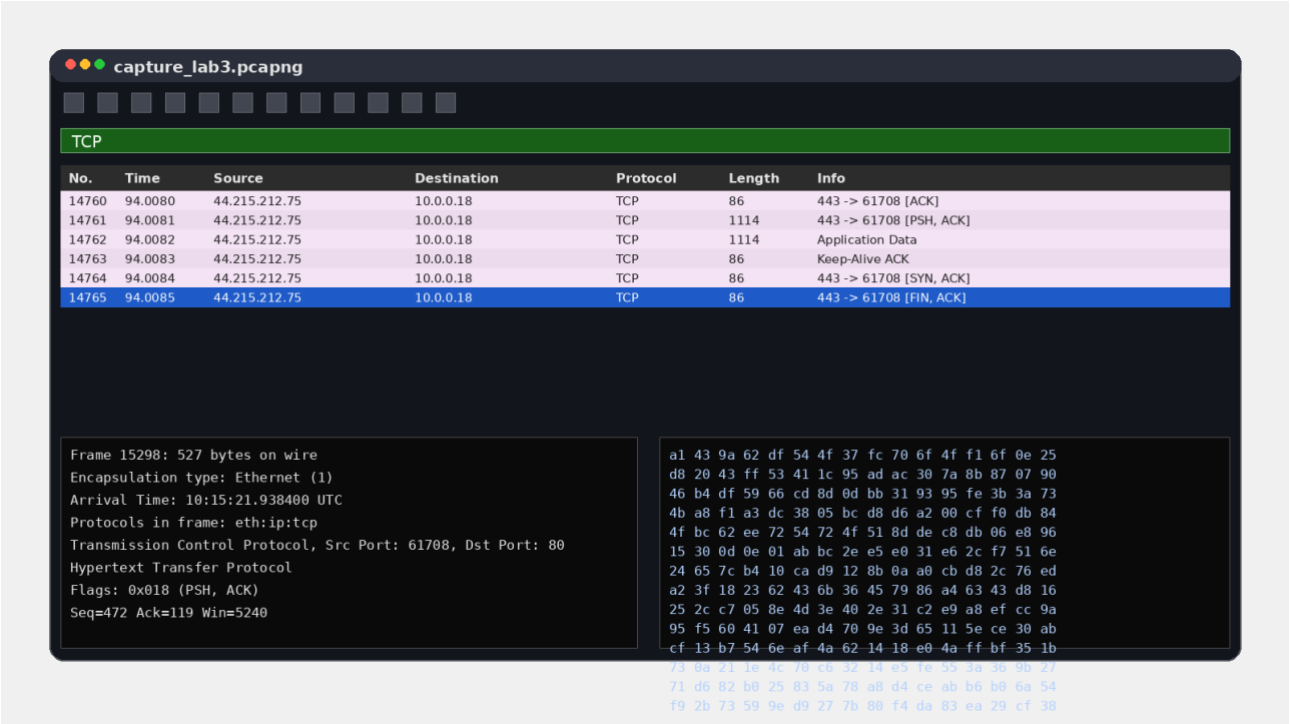

Figure 4. TCP/TLS packets associated with the same browsing session.

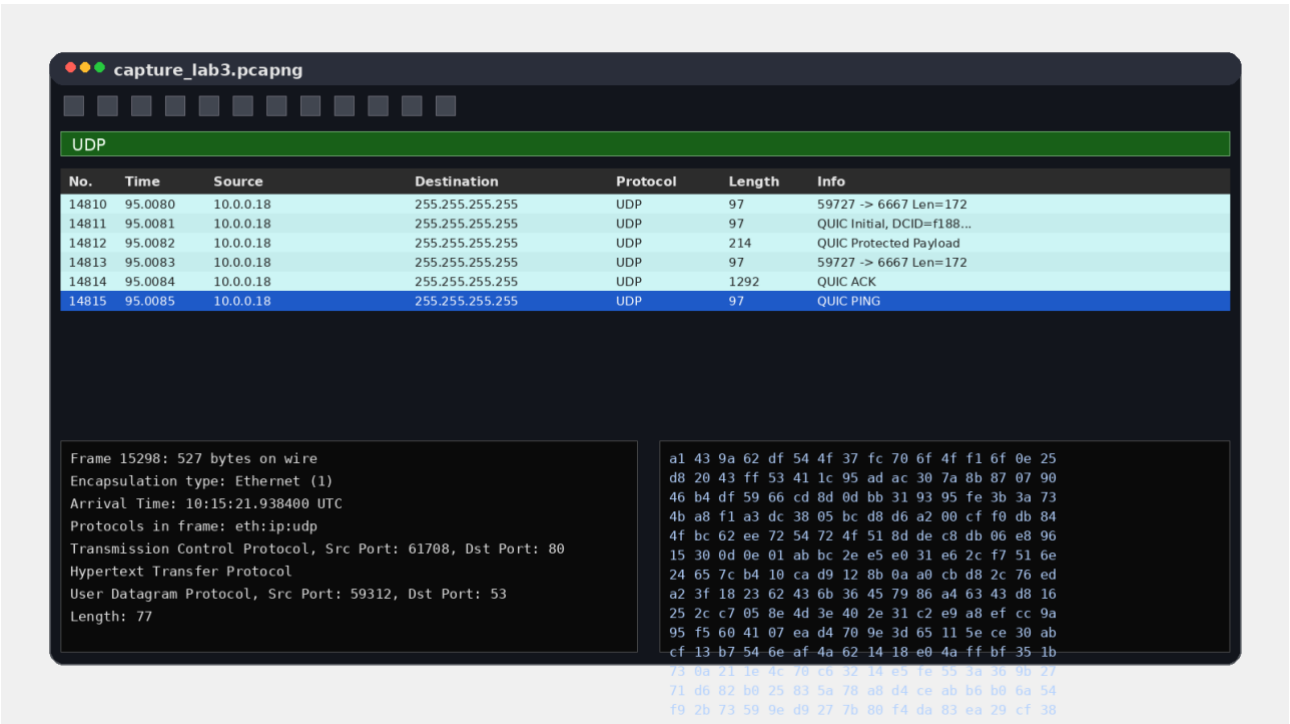

Figure 5. UDP and QUIC packets that appeared alongside DNS traffic.

Part 3: HTTP Request Summary

The following table summarizes sample requests identified from the filtered capture. All values below are fabricated for benchmarking purposes but follow realistic web-traffic patterns.

Method	Host	URI	Status	Payload (bytes)
GET	portal.example.edu	/portal/index	200 OK	527
GET	portal.example.edu	/css/app.css	200 OK	424
GET	cdn.example.edu	/img/banner.png	304 Not Modified	60
POST	api.example.edu	/api/login	302 Redirect	612

Part 4: Delay and Jitter

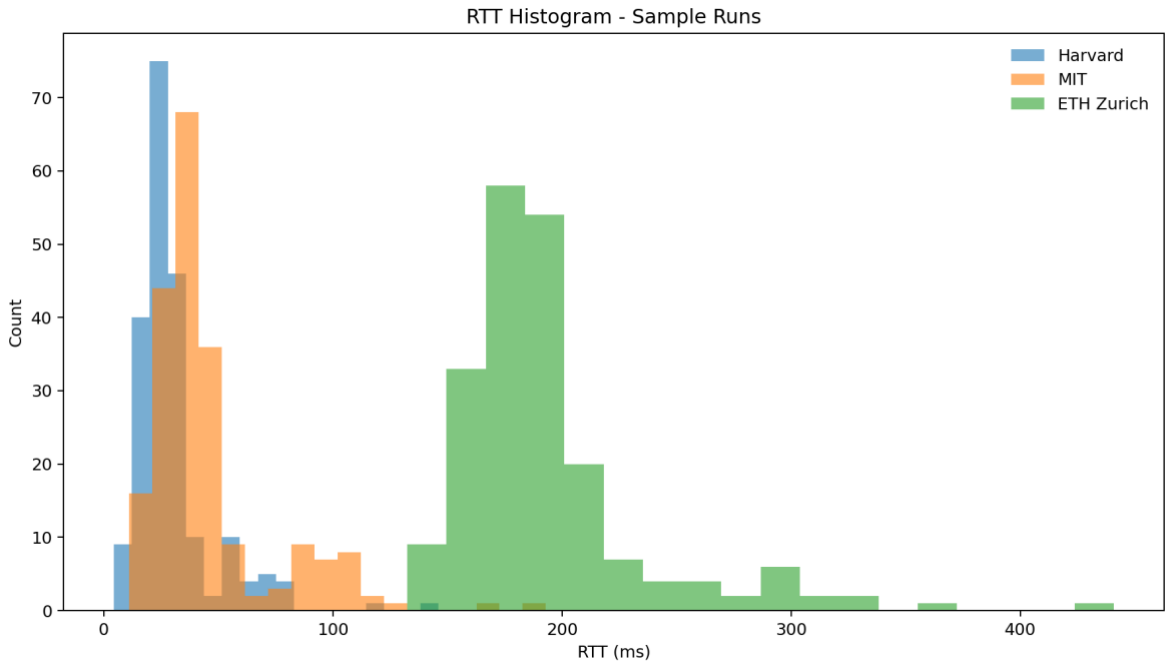

Figure 6. RTT histogram from three fabricated destination groups.

Host	Runs	Median RTT (ms)	P95 RTT (ms)	Std Dev (ms)	MAD Successive (ms)
harvard.edu	10	21.7	98.2	27.3	18.4
mit.edu	10	36.1	129.3	35.8	24.8
ethz.ch	10	177.2	267.9	37.4	34.0

Part 5: TCP 3-Way Handshake and Observations

A single connection was isolated and the time display was changed to “Seconds Since Previous Displayed Packet.” The sample handshake timings are shown below.

Entry	Packet Flags	Delta Time (s)
1st	SYN	0.000000
2nd	SYN, ACK	0.024381
3rd	ACK	0.000241

1. Name resolution appeared before every application flow, reinforcing that DNS latency is on the critical path.
2. The TCP and TLS filters highlighted the same sessions that later carried HTTP payloads.
3. Domestic destinations showed tighter RTT distributions than overseas destinations, while long-tail spikes were more visible in the remote host sample.
4. Custom columns for ports, host, and URI made triage substantially faster when multiple protocols interleaved in the same capture.